

CY 556: Organic Spectroscopy

Course Code	CY 556
Course Name	Organic Spectroscopy
L-T-P-C	3-0-0-3
Prerequisites	Intended for : MSc Chemistry and PhD (any discipline)
Distribution	Elective Course
Approval	43rd BoA (Oct 2021)

Course Contents

- **UV Spectroscopy**
Introduction, Principle of UV spectroscopy, Concept of chromophore, Solvent effect, Fluorescence and phosphorescence, Characteristic absorption of organic compounds, Woodward-Fieser rules for dienes and enones, Substituent effects, Model compound studies. [4 Lectures]
- **Infra-red (IR) Spectroscopy** Introduction, Molecular vibrations, factors influencing molecular frequencies, Infrared spectrometer, application of IR spectroscopy for identification of functional groups. [4 Lectures]
- **Mass Spectroscopy**
Basic principle, Ionization methods, Determination of molecular weight and molecular formula, Molecular ion and its recognition, Fragmentation and rearrangements, Examples of organic compounds from different classes such as hydrocarbon, hydroxyl compounds, ketones, aldehydes, carboxylic acids and esters, lactones, amines and amides, nitro compounds, nitriles and heteroaromatic compounds. [4 Lectures]
- **¹H NMR Spectroscopy**
The NMR Phenomenon, Theory of nuclear magnetic resonance, Chemical shift and factors affecting chemical shift, integral and integration in proton NMR, Chemical equivalence and magnetic equivalence, First order and second order spectra, Spin-spin coupling, Pascals triangle, Coupling constant, Factors influencing coupling constant, A2, AB, and AX spin systems, AMX, ABX and ABC spin system with coupling constants, D2O exchange, Shift reagents, Effect of chiral center, ¹⁹F and ³¹P NMR, Structure elucidation of organic compound using ¹H NMR. [12 Lectures]
- **¹³C NMR Spectroscopy and 2-D NMR spectroscopy**
Introduction, Correlation chart for ¹³C chemical shift, Proton-coupled ¹³C spectra, Proton-decoupled ¹³C spectra, carbon-deuterium coupling, NOE effects, Structural applications of ¹³C NMR, Fundamentals and applications of DEPT technique in NMR spectroscopy, Application of 2-D NMR spectroscopic techniques such as ¹H-¹H COSY, ¹H-¹³C COSY, HMBC and HSQC for structure determination of complex organic compounds. [10 Lectures]
- **Structural Elucidation of Organic Compounds**
Interpretation of spectroscopic data of unknown compounds, Application of UV- Vis, MS, IR and NMR spectroscopic techniques for solving structure of organic molecules [8 Lectures]

Text Books

1. Harald Gunther, **NMR Spectroscopy**, 2nd Edition, Wiley Publishers.
2. Jeremy K.M. Sanders, Edwin C. Constable, Brian K. Hunter and Clive M. Pearce, **Modern NMR Spectroscopy: A workbook of Chemical Problems**, 2nd Edition, Oxford University Press.
3. Jeffrey H. Simpson, **Organic Structure Determination, using 2-D NMR Spectroscopy, a problem- based approach**.

References

1. Erno Pretsch, Philippe Buhlmann, Martin Badertscher, **Structure Determination of Organic Compounds**, Tables of Spectral Data, Springer.
2. Robert M. Silverstein, Francis X. Webster, David Kiemle, **Spectrometric Identification of Organic Compounds**, 7th Edition, John Wiley & Sons.
3. William Kemp, **Organic Spectroscopy**, 3rd Edition, Palgrave Publishers Ltd.